

sm-lit-protocol

Literature rapid review protocol

Title: Literature Rapid Review Of Total Opioid Burden Metric Methodology And Reporting Within Preoperative Care Research.

Registration: The rapid review protocol is pre-registered on the [Open Science Framework](<https://doi.org/10.17605/OSF.IO/EFJ8G>)

Authors, contributions and support as per main manuscript.

Introduction

Rationale

There are a wide range of different opioid used across the range of medical specialities and patient groups, each with their own potency to provide analgesia. Opioid conversion values have offered the ability to convert a specific opioid into the equivalent units of a reference opioid, usually morphine, to produce doses in oral morphine equivalent milligrams (OME mg) or milligram morphine equivalent (MME).

A number of conversion value tables have been published in an attempt to provide a best estimate of the conversion values between opioids. Within clinical practice, these cited value sets are a good surrogate for the true conversion value when used in opioid rotation⁵. Despite this, application of the value sets remains variable in both clinical and research practice.

It is unclear how well the methodology is described and reported within research.

It is expec

Objectives

Primary research question: - Describe the methodologies of reporting of TOB metrics used in the perioperative literature.

Secondary research question(s)

- What proportion of study report the opioid proportions between comparative groups
- What proportion go study report the conversion values used to calculate the TOB metrics.
- To extract and calculate the outcomes de-novo for studies that publish the required minimum data of mean opioid doses within groups. ### Methods

The eligibility criteria is set using the PCCO framework as described in Table 1. Only papers published in 2025 will be considered to ensure the methodology reviewed is representative of current practice.

```
tbl_screening <- data.frame("Criteria" = c("Population", "Concept", "Context", "Other"),  
                           "Include" = c("All patient groups undergoing a surgical procedure",  
                                         "Patients not undergoing surgical procedures. Non-I"),  
                           "Exclude" = c("Patients not undergoing surgical procedures. Non-I"),  
                           "Outcome" = c("Opioid doses", "Conversion values", "Outcomes de-novo"),  
                           "Study" = c("Studies using a total opioid burden metric as a primary or  
                                         secondary outcome (Such as oral morphine equivalence (OME),  
                                         milligram morphine equivalence (MME) or opioid equivalence)",  
                                         "Patients undergoing surgical procedures",  
                                         "Any studies comparing a control against and intervention or  
                                         two interventions."))  
knitr::kable(tbl_screening)
```

Table 1: Litration extraction proforma

Cri- te- ria	Include	Exclude
Pop- ula- tion	All patient groups undergoing a surgical procedure	Patients not undergoing surgical procedures. Non-human studies
Con- cept	Studies using a total opioid burden metric as a primary or secondary outcome (Such as oral morphine equivalence (OME), milligram morphine equivalence (MME) or opioid equivalence)	Outcomes not including opioid totals
Con- text	Patients undergoing surgical procedures	Palliative care studies or Pharmacology studies
Other	Any studies comparing a control against and intervention or two interventions.	Studies with no difference between intervention groups

Search strategy

Databases searched will include PUBMED, SCOPUS and Web of Science through their respective web portals using the respective search string. No gray literature will be included as the review aims to capture the practice of current and published research methodologies.

i PUBMED search term

(((((("opioid equivalent" OR "oral morphine equivalence" OR "oral morphine equivalent" OR "oral morphine equivalents" OR "OME" OR "morphine milligram equivalents" OR "morphine milligram equivalence" OR "MME" OR "morphine equivalent daily dose" OR "MEDD") AND ("perioperative" OR "peri-operative" OR "postoperative" OR "post-operative" OR "surgical" OR "surgery" OR "post-surgical" OR "postsurgical")))) AND ("2025/01/01"[Date - Publication] : "2025/12/31"[Date - Publication]))) AND ("clinical trial"[Filter] OR "clinical trial" OR "controlled clinical trial" OR "randomized controlled trial" OR "interventional" OR "intervention study" OR "clinical trials as topic"[MeSH Terms] OR "cohort studies"[MeSH Terms] OR "cohort" OR "prospective" OR "retrospective" OR "case-control" OR "comparative study" OR "evaluation study"))

i SCOPUS search term

TITLE-ABS-KEY ("opioid equivalent" OR "oral morphine equivalence" OR "oral morphine equivalent" OR "oral morphine equivalents" OR "OME" OR "morphine milligram equivalents" OR "morphine milligram equivalence" OR "MME" OR "morphine equivalent daily dose" OR "MEDD") AND TITLE-ABS-KEY ("perioperative" OR "peri-operative" OR "postoperative" OR "post-operative" OR "surgical" OR "surgery" OR "post-surgical" OR "postsurgical") AND TITLE-ABS-KEY ("clinical trial" OR "controlled clinical trial" OR "randomized controlled trial" OR "interventional" OR "intervention study" OR "cohort" OR "prospective" OR "retrospective" OR "case-control" OR "comparative study" OR "evaluation study") AND PUBYEAR = 2025

i Web of Science search term

TS=(“opioid equivalent” OR “oral morphine equivalence” OR “oral morphine equivalent” OR “oral morphine equivalents” OR “OME” OR “morphine milligram equivalents” OR “morphine milligram equivalence” OR “MME” OR “morphine equivalent daily dose” OR “MEDD”) AND TS=(“perioperative” OR “peri-operative” OR “postoperative” OR “post-operative” OR “surgical” OR “surgery” OR “post-surgical” OR “postsurgical”) AND TS=(“clinical trial” OR “controlled clinical trial” OR “randomized controlled trial” OR “interventional” OR “intervention study” OR “cohort” OR “prospective” OR “retrospective” OR “case-control” OR “comparative study” OR “evaluation study”) AND PY=(2025)

Study records and screening

Data management

Extracts from each search term will be uploaded to the Covidence review platform. Duplicate items will be removed. At each stage, records will be extracted and stored for backup.

Selection process

Screening of title and abstract will happen within the Covidence platform screening section. It will be performed by one reviewer with a subset review of 20% being reviewed by a second author to reduce bias and verify quality. Any discrepancies will be reviewed and resolved through a moderation discussion.

Data collection process

Data collection will occur in two phases

1. **Full text screening and primary data extraction** - Manuscripts that pass screening will have a full text review. If at this stage it is not eligible it will be removed, if it is eligible it will have data extracted. Primary data collection will then occur by a single reviewer with a subset review of 20% being reviewed by a second author to reduce bias and verify quality
2. **Secondary data extraction** - A second stage data extraction will be performed for papers that cite/describe either a conversion value or report the opioid doses within control and intervention groups. This will provide further details, for that papers that do report their methodology. This will then occur by a single reviewer with a subset review of 20% being reviewed by a second author to reduce bias and verify quality.

Data items

Standard metadata including author, title, data of publication, journal of publication, doi and abstract (where possible) will be extracted at the search stage.

Primary data extraction: Review specific data will be extracted from all papers that pass screening using the proforma as detailed in Table 2.

```
nl      <- if (is_latex_output()) "\\\\" else "<br>"
bullet <- if (is_latex_output()) "\\textbullet\ " else "&bull; "

bullet_list <- function(x) paste0(bullet, x, collapse = nl)

tbl_extraction_form <- data.frame(
  "Variable" = c("Study type", "Intervention type", "Conversion set", "Opioid proportions"),
  "Options" = c(
    bullet_list(c("Randomised controlled trial", "Cohort study", "Case controlled study", "Other")),
    bullet_list(c("Surgical", "Anaesthetic", "Analgesia", "Other")),
    bullet_list(c("Described", "Cited", "Not described")),
    bullet_list(c("Reported", "Not reported"))
  ),
  "Description" = c("What is the methodology of the study? If the study is not comparing an intervention, is the conversion set used to generate the results?",
    "In the manuscript, is the conversion set used to generate the results?",
    "What type of intervention is given. For trials comparing interventions, is the type of intervention described?",
    "Assess the results section, Is it possible to establish the effectiveness of the intervention?"
  )
)

knitr::kable(
  tbl_extraction_form,
  format = if (is_latex_output()) "latex" else "html",
  escape = FALSE
)
```

Secondary data extraction: The data will be collected using an excel spreadsheet with the following data item names and descriptors.

```
template_path <- "data/data_collection_template.xlsx"

sheet_names <- readxl::excel_sheets(template_path)

process_sheet <- function(sheet) {
  readxl::read_excel(template_path, sheet = sheet, col_names = FALSE, n_max = 2) %>%
    mutate(row_id = c("Data_item", "Description")) %>%

```

Table 2: Literature extraction proforma

Variable	Options
Study type • Cohort study • Case controlled study • Other	• Randomised controlled trial What is the methodology of the study? If the study is not comparing an intervention
Intervention type • Anaesthetic • Analgesia • Other	• Surgical In the manuscript, is the conversion set used to generate the TOB metric described
Conversion set • Cited • Not described	• Described What type of intervention is given. For trials comparing surgical technique, approach
Opioid proportions • Not reported	• Reported Assess the results section, Is it possible to establish the proportion, mean or total d

```

tidyr::pivot_longer(cols = -row_id) %>%
tidyr::pivot_wider(names_from = row_id, values_from = value) %>%
mutate(Sheet = sheet) %>%
select(Sheet, `Data item` = Data_item, Description)
}

final_table <- purrr::map_dfr(sheet_names, process_sheet)

```

New names:

New names:

New names:

```

* `` -> `...1`
* `` -> `...2`
* `` -> `...3`
* `` -> `...4`
* `` -> `...5`
* `` -> `...6`
* `` -> `...7`
* `` -> `...8`
* `` -> `...9`
* `` -> `...10`
* `` -> `...11`
* `` -> `...12`
* `` -> `...13`

```

```
* `` -> `...14`
* `` -> `...15`
* `` -> `...16`
* `` -> `...17`
* `` -> `...18`
* `` -> `...19`
* `` -> `...20`
* `` -> `...21`
* `` -> `...22`
* `` -> `...23`
* `` -> `...24`
* `` -> `...25`
* `` -> `...26`
* `` -> `...27`
```

```
final_table %>%
  gt()
```

Outcomes

Primary outcomes:

- To report the proportion of manuscripts that use a total opioid burden metrics in perioperative research that report the conversion values used and the opioid doses within groups.

Secondary outcomes:

- To report the range and variability of the opioid conversion values used within the perioperative literature
- To report the variability in the reported outcomes and de-novo calculates outcome for studies that report the opioid doses within groups

Data synthesis

Data will be analysed quantitative. For papers that are eligible and have primary data extraction, the counts and proportions will be presented.

No qualitative analysis will be performed.

Sheet	Data item	Description
conversion_values	cite_ref_number	[autoassigned] incremented id number for the citation
conversion_values	cite_ref_text	[autoassigned] The copied citation from the reference
conversion_values	zotero_cite_key	[autoassigned] The zotero citation key of the reference
conversion_values	year	Year of publication
conversion_values	sources	If described, what sources or citations does the article use
conversion_values	format	What is the methods of generating the conversion value
conversion_values	provides_value_set	Does the article provide explicit conversion values
conversion_values	aim	If explicitly stated, what is the purpose of the study
conversion_values	reference	If explicitly stated, what is the reference drug
conversion_values	morphine_oral	Conversion value stated (add further opioid if reported)
conversion_values	morphine_iv	Conversion value stated (add further opioid if reported)
conversion_values	morphine_sc	Conversion value stated (add further opioid if reported)
conversion_values	morphine_epidural	Conversion value stated (add further opioid if reported)
conversion_values	morphine_intrathecal	Conversion value stated (add further opioid if reported)
conversion_values	oxycodone_oral	Conversion value stated (add further opioid if reported)
conversion_values	methadone_oral	Conversion value stated (add further opioid if reported)
conversion_values	fentanyl_iv	Conversion value stated (add further opioid if reported)
conversion_values	fentanyl_sc	Conversion value stated (add further opioid if reported)
conversion_values	remifentanyl_iv	Conversion value stated (add further opioid if reported)
conversion_values	sufentanyl_iv	Conversion value stated (add further opioid if reported)
conversion_values	fentanyl_transdermal	Conversion value stated (add further opioid if reported)
conversion_values	codeine_oral	Conversion value stated (add further opioid if reported)
conversion_values	hydrocodone	Conversion value stated (add further opioid if reported)
conversion_values	hydromorphone	Conversion value stated (add further opioid if reported)
conversion_values	oxymorphone	Conversion value stated (add further opioid if reported)
conversion_values	tapentadol	Conversion value stated (add further opioid if reported)
conversion_values	tramadol	Conversion value stated (add further opioid if reported)
study_summary	covidence_ref	[autoassigned] The reference number of the study
study_summary	authors	[autoassigned] The reference name of the study
study_summary	control_group	Text description of the treatment in the control group
study_summary	control_group_size	Number of patients within the control group
study_summary	intervention_group	Text description of the treatment in the intervention group
study_summary	NA	Number of patients within the intervention group
study_summary	population	Text description of the eligible participants
study_summary	note	Any notes with regards to the data collection
study_summary	reported_control_group_mean	Reported numerical value of the control group mean
study_summary	reported_intervention_group_mean	Reported numerical value of the intervention group mean
study_summary	reported_p_val	reported p value of the test of significant difference
study_doses	covidence_ref	[autoassigned] The reference number of the study
study_doses	authors	[autoassigned] The reference name of the study
study_doses	opioid	name of the opioid
study_doses	control_mean_dose	If reported, the mean dose of the opioid in the control group
study_doses	control_sd	If reported, the standard deviation of the dose of the opioid in the control group
study_doses	control_median_dose ⁸	If reported, the median dose of the opioid in the control group
study_doses	control_q1	If reported, the first quartile of the dose of the opioid in the control group
study_doses	control_q3	If reported, the third quartile of the dose of the opioid in the control group
study_doses	intervention_mean_dose	If reported, the mean dose of the opioid in the intervention group
study_doses	intervention_sd	If reported, the standard deviation of the dose of the opioid in the intervention group
study_doses	intervention_median_dose	If reported, the median dose of the opioid in the intervention group
study_doses	intervention_q1	If reported, the first quartile of the dose of the opioid in the intervention group

Subgroup analysis

For papers that provide a conversion value or the papers cited, the conversion values for each opioid will be presented with the range, mean and median.

For papers that provide opioid proportions, the reported outcome will be presented alongside the de-novo point estimate and the range estimate utilising the range presented previously.

Meta-bias

No planned assessment of meta-biases

Confidence in cumulative evidence

This is a rapid review of the methodology reporting, it aims to look at all published studies regardless of methodologies or study design therefore the body of evidence will not be assessed for strength.